

1. Identify the type of the following variables (ordinal, Poisson, binomial, continuous, etc.) (5 pts each)
 1. Responses on a survey ranking the quality of service on a scale of 1-10 **Ordinal**
 2. Dice rolls **Discrete**
 3. Pregnancy test result **Binomial**
 4. Religious affiliation **Nominal**
 5. Likelihood of being bitten by a dog in NYC **Poisson**
2. Explain the omnibus null. (5 pts)

The omnibus null hypothesis is a type of hypothesis that has multiple null hypotheses within a single test. It suggests that none of the several groups, factors, or situations under comparison show a significant difference. An example of the omnibus null being used is when using an ANOVA test for comparing different means. The omnibus null hypothesis states that there are no differences in means among the groups being compared. This means that all the group means are equal. When the omnibus null hypothesis is rejected, it will indicate that one or more of the individual null hypotheses under test are wrong, although it does not say which one is/are false.

3. What test would you use to compare the average height of a random sample of American men to the average height of a random sample of Swedish men? (5 pts)

To compare the average heights of both groups, you would use a two-sample unpaired t-test. This test is appropriate since we are testing two distinct groups (American men and Swedish men) and we want to compare their means on a continuous outcome variable (height). Each group represents a different set of participants. Participants in one group are unrelated to those in the other group. This test assesses whether the means of the two groups are significantly different from each other.

4. What test would you use to compare blood glucose levels before and after administration of a diabetes drug? (5 pts)

I would use a paired samples t-test to compare blood glucose levels before and after the administration of a diabetes drug. We would use this type of test since we want to compare their means on a continuous outcome variable and because we are comparing two related groups. Both groups consist of the same participants or subjects measured under different conditions or at different time points. Each participant's data is paired with their own data in the other condition.

5. A random sample of registered voters shows that being a registered Democrat has a high correlation with years of education. Can I conclude that education makes people be Democrats? Why or why not? (5 pts)

Correlation does not imply causation. It is not proven that education makes someone Democratic, even though there may be a strong link between having more years of school and being a registered Democrat. Geographical location, cultural background, and socioeconomic status are a few other variables that could have an independent impact on political affiliation and educational level. The observed association may be concluded by these third factors. Without taking other variables and other confounding variables into account, it is inappropriate to make causal assumptions regarding the relationship between education and political affiliation.

6. Why do we not simply run multiple t-tests to test multiple means? What test should we use instead? (5 pts)

An analysis of variance (ANOVA) test would be used in place of many t-tests. ANOVA allows us to account for the overall Type I error rate while concurrently comparing means across several groups. It evaluates if the means of three or more groups differ in a statistically significant way. When comparing multiple means with numerous t-tests, there is a greater chance of Type I error, or false positives. This happens because there is a greater possibility of producing at least one statistically significant result by chance alone the more tests that are performed.